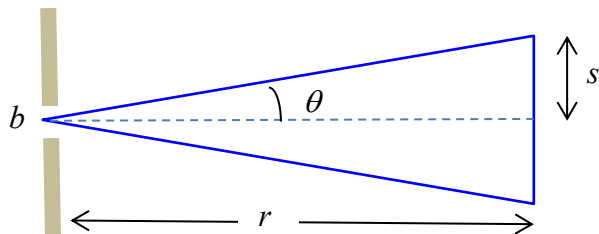


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Method 1: Using HUP,

➤ If assume $\Delta x = b$, then $r \leq 0.069m$

$$\Delta x \Delta p \geq h$$

$$(b) p \sin \theta \geq h \quad \text{where} \quad \sin \theta = \frac{\Delta p}{p}, \quad \sin \theta = \frac{s}{r}$$

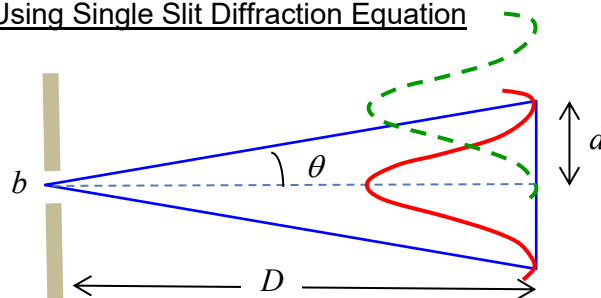
$$(b) mv \left(\frac{s}{r} \right) \geq h$$

$$(10^{-10}) (9.11 \times 10^{-31}) (10^6) \frac{0.5}{r} \geq h$$

$$r \leq 0.069m$$

➤ If assume $\Delta x = \frac{b}{2}$, then $r \leq 0.035m$

Method 2: Using Single Slit Diffraction Equation



For single slit diffraction, $\sin \theta = m \frac{\lambda}{b}$ where $m = 1, 2, 3, ..$ (1)

By Rayleigh Criterion, $\tan \theta \approx \frac{d}{D}$ (2)

For small angles, $\frac{d}{D} \approx \frac{\lambda}{b}$ where $p = \frac{h}{\lambda}$

$$\approx \frac{h/p}{b}$$

$$\frac{0.5}{D} \approx \frac{h / (9.11 \times 10^{-31}) (10^6)}{10^{-10}}$$

$$r \approx 0.069m$$